Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same

1 Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same

Study focus: Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000134 · EV0000135

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials that is polymers and this will be the last class of this module.

Key details

- In this class we are going to talk about molecular weight of polymers.
- That is number average molecular weight, weight average molecular weight, and viscosity average molecular weight.
- We know that its molecular weight is eighteen, and irrespective of the source from where we are going to take the water, the molecular weight is going to still be eighteen, and each of the molecule is going to have a molecular weight of 18, but the case of polymers is different because in a sample the molecular weight of the polymers, the chain length of the polymers is different.
- Each unit may have a different.
- Therefore, whenever we talk about molecular weight of a polymer, we talk about an average molecular weight.
- So this molecular weight now we know is a very important characteristic of a polymer sample because several properties of polymers are found to depend on the molecular weight.

- The entire range of mechanical properties which are very useful to determine the application of a polymer like stress strain, impact, fracture, fatigue, creep, stress relaxation, and cracking, they are all influenced by the average molecular weight and also the molecular weight distribution.
- There could be number average molecular weight, we can also have a weight average molecular weight and also viscosity average molecular weight.
- Number average molecular weight is denoted by Mn bar.
- The bar over M stands for the average.
- with the total number of polymer units present.
- Then the total weight would be obviously n1 multiplied by m1.
- Summation of NIMI divided by summation of NI would give me the number average molecular weight.
- Of course the summation is over I and what is I?
- It is the particular species we are talking about.
- So I will be 1 for the first species, so N1, M1.
- and divided by N1.
- And similarly, N2, M2.
- And then divide by the total number of units that is n1 plus n two plus n three and so on so forth, what we saw here.
- This is just writing it in a short form.
- Depend only on the number of the species and not on the size or the nature of the species.
- And such colligative properties you are already aware, like osmotic pressure, elevation and boiling point, depression and freezing point, or even relative lowering of vapor pressure.
- So all these could be used to determine the number average molecular weight because polligative properties depend only on number and when we are talking of number average molecular weight we are only concerned about the number of polymer units.

Example or application from the lecture

- When we talk of molecular weight of any compound, for example, water.
- Let us look at each one of them.
- Suppose N1, N2, N3 are the number of polymer units having molecular weights of N1, M2, N3.

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Core explanation · 05:00–09:59

use the mass we are going to consider the mass of each of these species and not just the number.

Key details

- Why many times the number average molecular weight does not give you a clear picture of the average molecular weight.
- We want to we understand that the higher masses would contribute more to the molecular weight than the lower mass.

- In number average molecular weight here we are talking of the mass.
- So therefore mass average molecular weight on weight average molecular weight will be.
- M1 multiplied by capital M1 plus M2 multiplied by capital M2 and so on divided by the masses M1 plus M2 plus so on.
- Again, they are masses of the polymer units having molecular weights M1, M2, M3 respectively.
- M1 is replaced by N1 M1.
- So therefore it becomes N1 M1 squared.
- And in the denominator, we had M1 plus M2 plus M3.
- plus n two n two and so on.
- I stands for the that particular species.
- Obviously some experiment which depends on the size like how we chose polligative properties for.
- Number average molecular weight because colligative properties depend only on number.
- Here we are going to look for an experiment which is going to be dependent on the size, and these would be sedimentation velocity or light scattering.
- Sedimentation velocity will depend on the weight of the units and light scattering would depend on the size of the each of these units and therefore these two experiments could be used to determine weight average molecular weight.
- is known as the polydispersity index.
- This is known as a poly dispersity index.
- What does this tell us?
- It is a measure of the broadness of a molecular weight distribution of a polymer, that is, the larger the polydispersity index, the broader is the molecular weight distribution.
- Now if you had a mono dispersed polymer that means only one type of polymer unit.
- From the formula we can understand that weight average will become equal to the number average.
- How much is the weight average more than the number average?
- That is what we are talking about, the ratio.
- Weight average by the number average, how much is it different from one?
- That will give us the polydispersity index, meaning that if it was equal.
- If it all the polymer chains were of equivalent or very close to equal length, then we would have had PDI that is a polydispersity index is equal to one.

Example or application from the lecture

- There we consider number.

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Core explanation · 10:00–14:59

But if they are different, then the ratio m w bar by mn bar, that is a poly dispersed index would be greater than one.

Key details

- Now looking at this picture here we see that these polymer chains there's very narrow dispersity, so therefore we can that is how we when we calculate the poly dispersity index, this will come very close to one, whereas here we see the polymer chains are of different lengths.
- This is known as the mark hoving equation.
- It is the same which is used in the Markov which comes in the Markoving.
- equation and this is how we determine using viscosity experimentally we can.
- Determine the viscosity average molecular weight.
- And this value of A is going to be dependent on the polymer solvent system and also the temperature.
- And usually the value lies.
- between 0.5 and 0.9.
- Because we know that larger molecules will contribute more to viscosity.
- And we found that weight average is always higher than the number average molecular weight.
- This peak here, when we get a bell shaped curve like this, the peak will represent the number average molecular weight, and since this is the molecular weight scale, we find weight average is always more than the number average molecular weight.
- B so that is the reason that reason I explained just now that because viscosity.
- The larger molecules contribute more to the viscosity of the system and therefore viscosity average molecular weight will be closer to the weight average molecular weight.
- number average molecular weight.
- So with that we come to the end of module seven and in the next class learn about organic light emitting diodes.

Example or application from the lecture

- the molecular weight of each of the species and the number of polymer units of that particular species, then the equation is given this.
- Viscosity average molecular weight is given by summation of NIMI one plus A divided by summation of NIMI and the whole thing is raised to the power one by A.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9955; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00036	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice